



School of Computer Science and Statistics

Week 6 Assignment

CS7CS2 – Optimisation for Machine Learning

Lecturer: Dr. Mouli Chakraborty

Contact: mchakrab@tcd.ie

General Guidelines

- You may discuss ideas with classmates; however, **all written work and code must be entirely your own**. Do not share code. All submissions will be checked for plagiarism.
- Submit **one PDF report** and **one .ipynb or .py** via Blackboard (no zip files).
- Marks are awarded primarily for explanation, reasoning, and interpretation (not raw code/output).
- All plots must have labelled axes and a brief explanation of what they show.

Submission Rules

Submit:

- A short PDF report with codes, plots and brief explanations (within 20 pages).
 - A **.ipynb** or **.py** file containing the executable Python code.
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Q1:

Consider the linear regression $\hat{y} = X\theta$ with $\theta \in \mathbb{R}^2$ and

$$L(\theta) = \frac{1}{2m} \|X\theta - y\|_2^2.$$

Generate synthetic data: $m = 1000$, $X_{ij} \sim \mathcal{N}(0, 1)$, $\theta^* = [3, 4]^T$, $y = X\theta^* + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 1$, initialise $\theta_0 = [1, 1]^T$.

(a) Full-batch GD : Run 80 iterations with constant step size (e.g. $\alpha = 0.5$).

Plot: (i) $L(\theta_t)$ vs iteration (log-scale), (ii) trajectory on contour plot of $L(\theta_1, \theta_2)$.

(b) Mini-batch SGD with different batch-size comparison: Shuffle once per epoch. Run 400 updates (same α as (a)). Compare $b = 5$ with $b = 20$.

Plot: (i) one loss plot with GD, SGD($b = 5$), SGD($b = 20$), (ii) one contour plot with all three trajectories.

(c) SGD noise (NAG vs Adagrad): Using $b = 10$ and 400 updates, implement NAG ($\beta = 0.9$) and Adagrad Plot: (i) loss plot comparing SGD (constant), NAG, Adagrad, (ii) contour plot with trajectories.

Q2:

Consider the Toy Neural Net model:

$$\hat{y}(u; x) = x_2 \tanh(x_1 u), \quad x = [x_1, x_2]^T, \quad J(x) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} \left(\hat{y}(u^{(i)}; x) - y^{(i)} \right)^2.$$

Generate data: $m = 1000$, $u^{(i)} \sim \text{Uniform}[-2, 2]$, $x^* = [1, 3]^T$, $y^{(i)} = \hat{y}(u^{(i)}; x^*) + \epsilon^{(i)}$, $\epsilon^{(i)} \sim \mathcal{N}(0, 0.05^2)$, initialise $x_0 = [1, 1]^T$.

- (a) Implement full-batch GD as baseline : Run 500 iterations with constant step size (e.g. $\alpha = 0.75$). Plot: (i) $J(x_t)$ vs iteration (log-scale), (ii) trajectory on contour plot of $J(x_1, x_2)$.
- (b) Implement Mini-batch SGD and compare with different batch sizes: Shuffle once per epoch. Run 500 updates (same α as (a)). Compare $b = 5$ and $b = 20$. Plot: (i) one loss plot with GD, SGD($b = 5$), SGD($b = 20$), (ii) one contour plot with all three trajectories.
- (c) SGD noise (NAG vs Adagrad): Using $b = 10$ and 500 updates, implement NAG ($\beta = 0.9$) and Adagrad. Plot: (i) loss plot comparing SGD (constant), NAG, Adagrad, (ii) contour plot with trajectories.

Q3:

Consider the Rosenbrock function:

$$f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2, \quad x_0 = [-1, 1]^T.$$

- (a) Use Gradient Descent as a baseline : Implement gradient descent for 200 iterations with step size $\alpha = 10^{-3}$. Produce a plot of $f(x_t)$ vs iteration (log-scale).
- (b) Compute the gradient and Hessian using finite differences, and implement Newton updates for 200 iterations with constant scaling factor (e.g. $\alpha = 0.7$). Produce a plot of $f(x_t)$ vs iteration (log-scale).
- (c) Damped Newton (line-search damping)
Implement a damped Newton method by reducing the step length if the objective does not decrease (e.g., shrink factor 0.5 until a decrease is obtained). Plot: (I) $f(x_t)$ vs iteration (log-scale) comparing GD, Newton, and damped Newton, (II) Contour plot of $f(x_1, x_2)$ with trajectories of GD (subsamped), Newton, and damped Newton.

Hints : Damped Newton (Backtracking):

Given $x^{(0)} = x_0$, parameters $\alpha_0 \in (0, 1]$, shrink factor $\rho \in (0, 1)$, maximum backtracking steps K , and total iterations T :

For $t = 0, \dots, T - 1$:

1. Compute $f_t = f(x^{(t)})$, $g_t = \nabla f(x^{(t)})$, $H_t = \nabla^2 f(x^{(t)})$.
2. Solve for Newton direction
 $(H_t + \lambda I)p_t = g_t$.
3. Set $\alpha = \alpha_0$. For $k = 1, \dots, K$: $x_{\text{new}} = x^{(t)} - \alpha p_t$. If $f(x_{\text{new}}) < f_t$, accept and break; else $\alpha \leftarrow \rho \alpha$.
4. If no step accepted, use fallback: $x_{\text{new}} = x^{(t)} - 10^{-4} g_t$.
5. Update $x^{(t+1)} = x_{\text{new}}$.

Return $\{x^{(t)}\}$ and $\{f(x^{(t)})\}$.
